

G H Patel College of Engineering & Technology
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Internal Theory Examination

Subject Code: 202044501**Total Marks: 20****Subject : Computer Networks****Academic Year: 2024 – 25****Semester: IV (Even)****Date: 28/02/2025****Time: 9:15 AM to 10:15 AM****Note:** Figures to the right indicates full marks**Q:1 Answer the following short questions:****[05]**

- 1) Define the terms: (a) Throughput (b) Propagation Delay

Ans: Throughput:

Throughput is the rate at which data is successfully transmitted over a network channel. It is usually measured in bits per second (bps), kilobits per second (Kbps), or megabits per second (Mbps). Throughput depends on factors like network bandwidth, latency, and congestion.

Propagation Delay:

Propagation delay refers to the time it takes for a signal to travel from the sender to the receiver over a given medium. It is calculated as the distance divided by the propagation speed of the signal in the medium (e.g., speed of light in fiber optic cables or electromagnetic waves in free space). It is usually measured in microseconds (μ s) or milliseconds (ms).

- 2) Which topology is most commonly used in modern Ethernet LANs?

Ans: Star Topology

- 3) What is the maximum channel utilization of Slotted ALOHA?

Ans: The maximum channel utilization of Slotted ALOHA is 36.8% or 0.368 ($\approx$ 37%).

- 4) What is the main advantage of packet switching over circuit switching?

Ans: Efficient bandwidth utilization, Dynamic Resource Allocation, Better Scalability, Fault Tolerance.

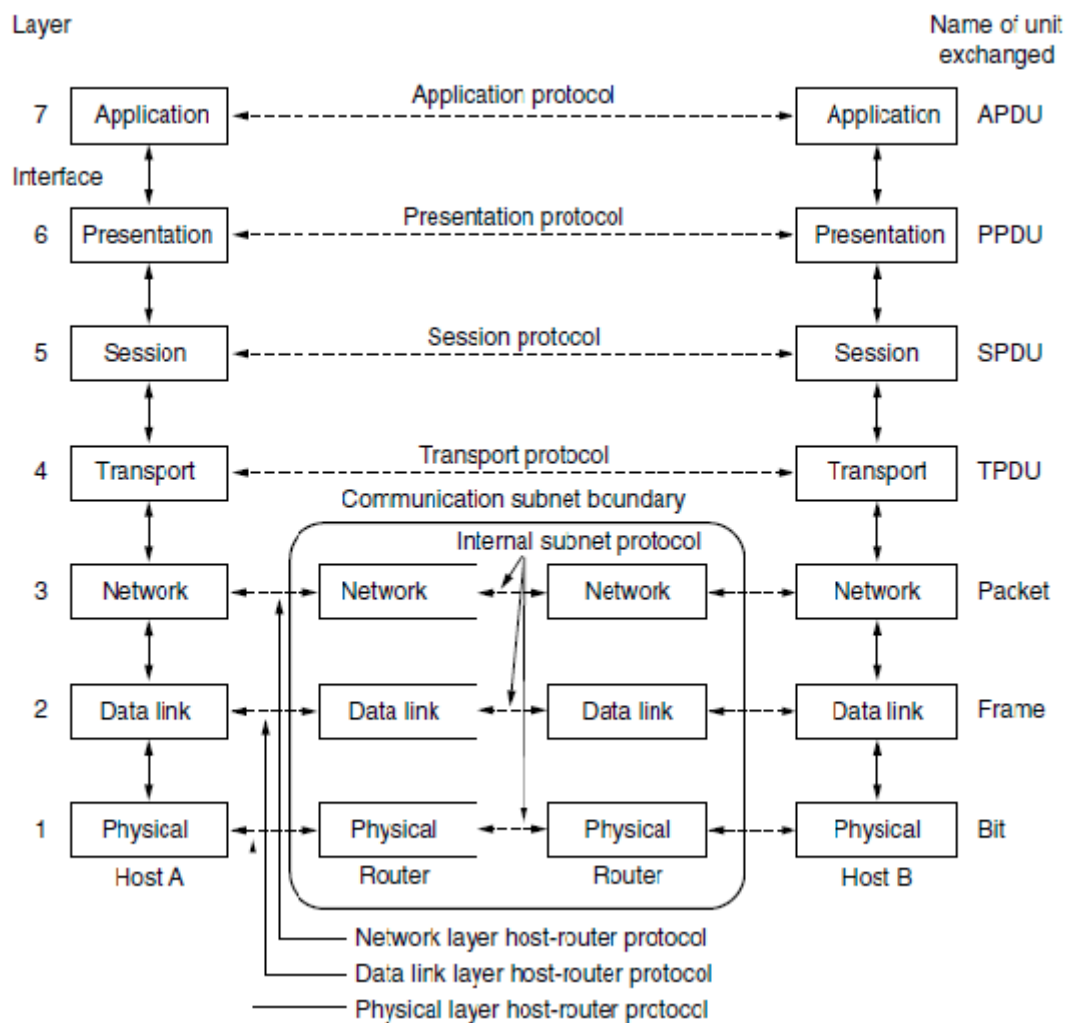
- 5) "Parity bits" are used for which of the following purposes?

- (a) Encryption of data
- (b) To transmit faster
- (c) To detect errors
- (d) To identify the user

Ans: (c) To detect errors

Q:2 Draw the layered architecture of OSI reference model and write at least two services provided by each layer of the model. **[05]**

Ans: OSI reference model:



Services Provided by Each Layer:

1. Physical Layer

- Transmission of raw bits over a physical medium.
- Signal encoding and modulation.

2. Data Link Layer

- Error detection and correction (e.g., CRC, checksum).
- MAC (Media Access Control) addressing for device identification.

3. Network Layer

- Logical addressing (e.g., IP addressing).
- Routing of data packets between different networks.

4. Transport Layer

- Reliable data transmission (e.g., TCP ensures delivery).
- Flow control to prevent data overflow.

5. Session Layer

- Establishment, maintenance, and termination of communication sessions.
- Synchronization of data exchange (e.g., checkpoints in data streams).

6. Presentation Layer

- Data encryption and decryption (e.g., SSL/TLS).
- Data compression (reducing file size for transmission).

7. Application Layer

- Provides services like file transfer (FTP) and email (SMTP, IMAP).
- Web browsing (HTTP/HTTPS).

Q:3 Explain Selective Repeat ARQ protocol and discuss how it is better than Go-Back_N [04] protocol.

Ans **Selective Repeat Automatic Repeat Request (ARQ)** is an error-control protocol used in reliable data transmission. It allows the sender to retransmit only the specific frames that were lost or received in error, rather than retransmitting all frames after an error (as in Go-Back-N ARQ).

Working of Selective Repeat ARQ:

1. Sliding Window Mechanism:

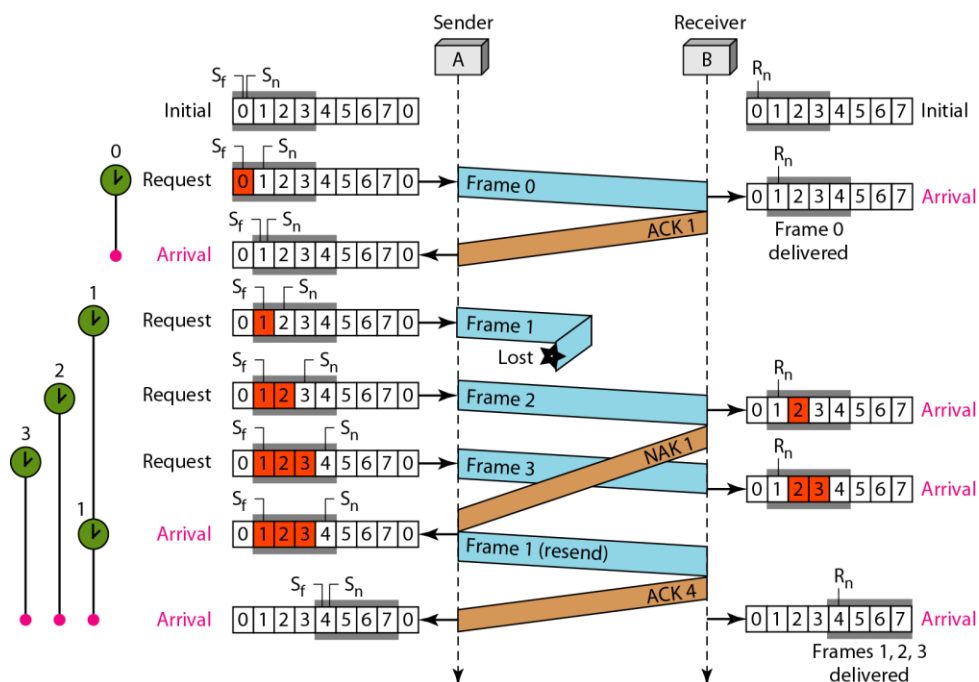
- Both sender and receiver maintain a **window** of frames that they can send or receive at a time.
- The sender keeps track of unacknowledged frames within the window.

2. Acknowledgment:

- The receiver individually acknowledges each correctly received frame.
- If an error occurs in a frame, the receiver **buffers correctly received frames** and requests only the erroneous frame to be retransmitted.

3. Retransmission:

- The sender **only retransmits** the frames that were lost or corrupted, instead of resending all frames from the point of error.



Why Selective Repeat is Better than Go-Back-N?

- **Reduces retransmission overhead** → Only retransmits erroneous frames.
- **Better bandwidth utilization** → Avoids unnecessary retransmission.
- **More suitable for high-latency networks** → As it minimizes retransmissions, it works well in long-distance communication.

Q:4 Attempt the following (any two)**[06]**

- 1) How the p-persistent is different from 1-persistent in CSMA/CD? Explain how the Backoff time is set in the case of collision.

Ans:

Difference Between p-Persistent and 1-Persistent CSMA/CD:

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is a network protocol used in Ethernet to avoid collisions during transmission. There are different types of CSMA/CD based on how a station decides to transmit:

Feature	p-Persistent CSMA	1-Persistent CSMA
Transmission Decision	The station transmits with probability p and defers with probability (1 - p) when the channel is idle.	The station transmits immediately when the channel is idle.
Collision Probability	Lower collision probability since not all stations transmit at the same time.	Higher collision probability, as multiple stations might send data simultaneously.
Used In	Suitable for time-slotted networks (e.g., wireless networks).	Used in unslotted Ethernet networks.
Efficiency	More efficient in networks with moderate to high traffic.	Less efficient due to more frequent collisions.

Backoff Time in CSMA/CD

When a **collision** occurs, CSMA/CD uses an **exponential backoff algorithm** to decide when to retransmit. The steps are:

1. Detection of Collision:

- If a collision is detected, all stations stop transmission immediately.
- They send a **jam signal** to ensure that all nodes are aware of the collision.

2. Binary Exponential Backoff (BEB) Algorithm:

- Each station waits for a random **backoff time** before attempting to retransmit.
- The backoff time is selected from a range of **0 to $(2^k - 1)$ time slots**, where **k** is the number of collision attempts (capped at 10).
- For example, after the first collision ($k = 1$), the station randomly picks a slot from **{0,1}**; after the second collision ($k = 2$), it picks from **{0,1,2,3}**, and so on.

3. Increasing Wait Time:

- If collisions continue, the waiting time increases exponentially.
- After **16 consecutive collisions**, the transmission attempt is **aborted**.

2) Explain the functionality of Repeater, HUB and Router.

Ans:

1. Repeater

Functionality:

- Receives a weak or corrupted signal and regenerates it before forwarding.
- Used in long-distance communication (e.g., fiber optics, coaxial cables).
- Does not filter traffic or prevent collisions.

2. HUB

Functionality:

- Works as a multi-port repeater by forwarding incoming signals to all ports.
- Cannot filter data; every device connected to the hub receives the same message.
- Increases network congestion as all devices share the same bandwidth.

3. Router

Functionality:

- Determines the best path for data packets to travel.
- Uses routing tables and protocols (e.g., OSPF, BGP) to manage traffic.
- Can provide firewall protection and NAT (Network Address Translation).
- Supports both wired and wireless communication (Wi-Fi routers).

3) Describe the guided transmission media.

Ans:

Guided Transmission Media

Guided transmission media refers to physical cables or wired communication channels that guide data signals from one device to another. These media provide a direct and confined path for data transmission.

Types of Guided Transmission Media:

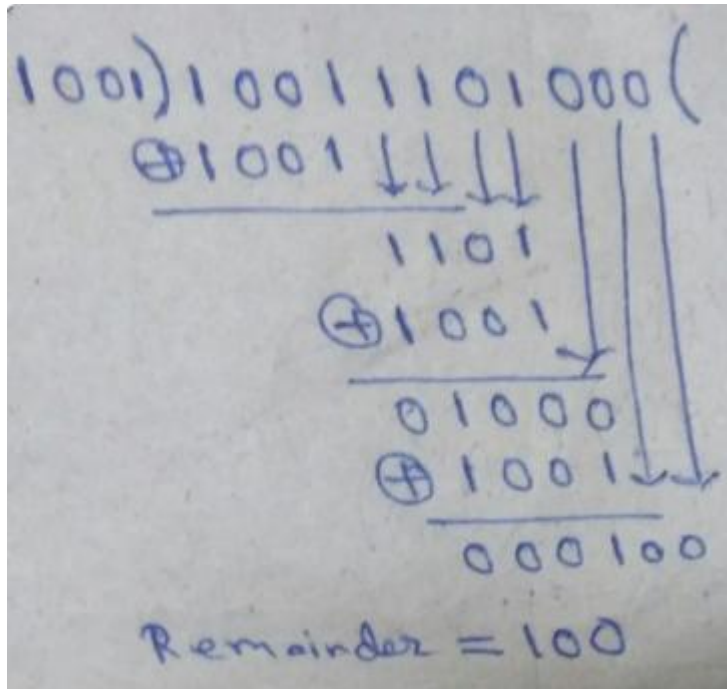
Medium	Description	Speed & Distance	Common Uses
Twisted Pair Cable	Two insulated copper wires twisted together to reduce interference.	Up to 10 Gbps , Distance: 100m (Ethernet).	Telephone lines, LANs, DSL connections.
Coaxial Cable	A central conductor with multiple layers of shielding to reduce signal loss.	Up to 10 Gbps , Distance: 500m .	Cable TV, broadband internet.

Fiber Optic Cable	Uses light signals for high-speed, long-distance communication.	Up to 100+ Tbps , Distance: 100 km+ .	High-speed internet, backbone networks.
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- 4) Bit stream 10011101 is to be transmitted using the standard CRC method with divisor value x^3+1 . Generate the CRC code word.

Ans: Generator polynomial= $x^3+1=1*x^3+0*x^2+0*x^1+1*x^0=1001$

So we've to append three 0s at the end of message



Message transmitted=10011101100

ALL THE BEST
